

# Antonin Sulc, PhD

Innovatrics · Czech Republic  
sulc.antonin@gmail.com · +420 731 250 375 · <http://sulcantonin.github.io>

## RESEARCH INTERESTS

---

Autonomous Systems (Particle Accelerators) · Computer Vision (Geometry, Light Fields) · Anomaly Detection · Physics-Informed Machine Learning · Agentic AI

## TECHNICAL SKILLS

---

- **Core AI/ML:** Anomaly Detection, LLMs (Fine-tuning, LoRA), Computer Vision.
- **Engineering & MLOps:** Python, Docker, Git, SQL, High Performance Computing (HPC), Linux envs.
- **Specialized:** Quantum Computing (Qiskit), Geometric Vision, Physics-Informed ML.
- **Languages:** English (C1), German (B2), Czech (Native).

## ACADEMIC APPOINTMENTS

---

|                                                                                                                                                                                                                                                                                                                       |                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>Innovatrics</b><br><i>Researcher</i>                                                                                                                                                                                                                                                                               | <b>Aug 2026 – Present</b><br><i>Prague, Czech Republic</i>  |
| <ul style="list-style-type: none"><li>• <b>Topic:</b> Intelligent algorithms for biometry</li></ul>                                                                                                                                                                                                                   |                                                             |
| <b>Lawrence Berkeley National Lab (LBNL)</b><br><i>Researcher</i>                                                                                                                                                                                                                                                     | <b>Jan 2025 – June 2026</b><br><i>Berkeley, USA</i>         |
| <ul style="list-style-type: none"><li>• <b>Topic:</b> Agentic AI for Accelerator Controls.</li><li>• Developing intelligent algorithms for autonomous accelerator operation.</li></ul>                                                                                                                                |                                                             |
| <b>Helmholtz Association (DESY/XFEL)</b><br><i>Senior Data Scientist</i>                                                                                                                                                                                                                                              | <b>May 2021 – Oct 2024</b><br><i>Hamburg, Germany</i>       |
| <ul style="list-style-type: none"><li>• <b>Topic:</b> Real-time accelerator control algorithms. <b>PIs:</b> Prof. Eichler, Dr. Wilksen.</li><li>• Developed unsupervised log anomaly detection and "PACuna" (fine-tuned LLMs).</li><li>• Researching "ChatQCD" and LLM applications in High Energy Physics.</li></ul> |                                                             |
| <b>University of Haifa - Marine Imaging Lab</b><br><i>Visiting Researcher</i>                                                                                                                                                                                                                                         | <b>Mar 2020 – Aug 2020</b><br><i>Haifa, Israel</i>          |
| <ul style="list-style-type: none"><li>• Developed algorithms for robust scene reconstruction and shape from refraction.</li></ul>                                                                                                                                                                                     |                                                             |
| <b>National Institute of Informatics (NII)</b><br><i>Visiting Researcher</i>                                                                                                                                                                                                                                          | <b>Oct 2018 – Mar 2019</b><br><i>Tokyo, Japan</i>           |
| <ul style="list-style-type: none"><li>• Research on spectral defocus blur and depth estimation.</li></ul>                                                                                                                                                                                                             |                                                             |
| <b>Vendavo Inc.</b><br><i>Software Engineer &amp; Data Scientist</i>                                                                                                                                                                                                                                                  | <b>Jan 2014 – Dec 2015</b><br><i>Prague, Czech Republic</i> |
| <ul style="list-style-type: none"><li>• Development of recommender systems for B2B pricing.</li></ul>                                                                                                                                                                                                                 |                                                             |

## EDUCATION

---

|                                                                                                                                                                                                      |                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>University of Konstanz</b><br><i>PhD in Artificial Intelligence (Computer Vision)</i>                                                                                                             | <b>2015 – 2020</b><br><i>Germany</i> |
| <ul style="list-style-type: none"><li>• Grade: <i>Magna Cum Laude</i> (1.0) · Advisor: Prof. Dr. Bastian Goldluecke</li><li>• Focus: Non-standard geometries, light-fields, photogrammetry</li></ul> |                                      |

## SOFTWARE & OPEN SOURCE

---

### **event2vector**

[PyPI](#) · [GitHub](#)

A scikit-learn style Python library providing a geometric approach to learning composable, highly interpretable representations of discrete event sequences.

### **torchmodal**

[PyPI](#) · [GitHub](#)

A framework introducing Differentiable Modal Logic for PyTorch—enabling the training of Modal Logical Neural Networks (MLNNs).

### **osprey-framework**

[PyPI](#) · [GitHub](#)

A scalable, production-ready framework for orchestrating agentic AI in safety-critical facility operations. Deployed at the Advanced Light Source.

### **fastlsq**

[PyPI](#) · [GitHub](#)

PDE solver and inverse problem under second

## SELECTED PUBLICATIONS

---

1. **A. Sulc**. Continuous Modal Logical Neural Networks: Modal Reasoning via Stochastic Accessibility *arXiv preprint*, under review, 2026.
2. **A. Sulc**. Differentiable Modal Logic for Multi-Agent Diagnosis, Orchestration and Communication. *arXiv preprint*, under review, 2026.
3. **A. Sulc**. FastLSQ: Solving PDEs in One Shot via Fourier Features with Exact Analytical Derivatives. *ICLR: AI & PDE*, 2026.
4. **A. Sulc**. Differentiable Logical Programming for Quantum Circuit Discovery and Optimization. *arXiv preprint*, under review, 2026.
5. T. Hellert, D. Bertwistle, S. C. Leemann, **A. Sulc**, M. Venturini. Agentic artificial intelligence for multistage physics experiments at a large-scale user facility particle accelerator. *Physical Review Research*, 2026.
6. T. Hellert, J. Montenegro, **A. Sulc**. Osprey: Production-ready agentic AI for safety-critical control systems. *APL Machine Learning*, 2026.
7. **A. Sulc**. Modal Logical Neural Networks. *arXiv preprint*, under review, 2025.
8. **A. Sulc**. Event2Vec: A Geometric Approach to Learning Composable Representations of Event Sequences. *NeurReps PMLR Proceedings*, 2025.
9. **A. Sulc**. Quantum Noise Tomography with Physics-Informed Neural Networks. *NeurIPS ML4Physics Workshop*, 2025.
10. **A. Sulc**, T. Hellert, A. Reed, A. Carpenter et al. eLog Analysis for Accelerators: Status and Future Outlook. *16th International Particle Accelerator Conference (IPAC)*, 2025.
11. **A. Sulc**, P. Connor. QCD in Language Models: What do they really know about QCD? *EPS-High Energy Physics*, 2025.
12. **A. Sulc**, P. Connor. ChatQCD: Let Large Language Models Explore QCD. *42nd ICHEP*, 2024.
13. **A. Sulc**, T. Hellert, R. Kammering, H. Hoschouer, J. St. John. Towards Agentic AI on Particle Accelerators. *NeurIPS ML4Physics Workshop*, 2024.
14. **A. Sulc**, G. Hartmann, J. Maldonado, V. Kain et al. Towards Unlocking Insights from Logbooks Using AI. *15th International Particle Accelerator Conference (IPAC)*, 2024.

15. **A. Sulc**, R. Kammering, A. Eichler, T. Wilksen. PACuna: Automated Fine-Tuning of Language Models for Particle Accelerators. *NeurIPS ML4Physics Workshop*, 2023.
16. **A. Sulc**, A. Eichler, T. Wilksen. Unsupervised Log Anomaly Detection with Few Unique Tokens. *IET Information Security*, 2023.
17. **A. Sulc**, A. Eichler, T. Wilksen. A data-driven anomaly detection on SRF cavities at the European XFEL. *Journal of Physics: Conference Series*, 2023.
18. S. Ishihara, **A. Sulc**, I. Sato. Depth Estimation Using Spectrally Varying Defocus Blur. *Journal of the Optical Society of America A*, 2021.
19. S. Ishihara, **A. Sulc**, I. Sato. Depth from Spectral Defocus Blur. *ICIP*, 2019.
20. O. Johannsen, **A. Sulc**, B. Goldluecke. What Sparse Light Field Coding Reveals About Scene Structure. *CVPR*, 2016.

## SERVICE & LEADERSHIP

---

- **Program Chair:** Foundations of Agentic Systems Theory (FAST) @ AAAI 2025.
- **Scientific Chair:** ICFA Beam Dynamics Mini-Workshop on ML for Particle Accelerators.
- **Reviewer:** CVPR, NeurIPS (2023-2025), BMVC, ICLR.